Technical Service Bulletin

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Revision to Procedure 001-026, Cylinder Block	

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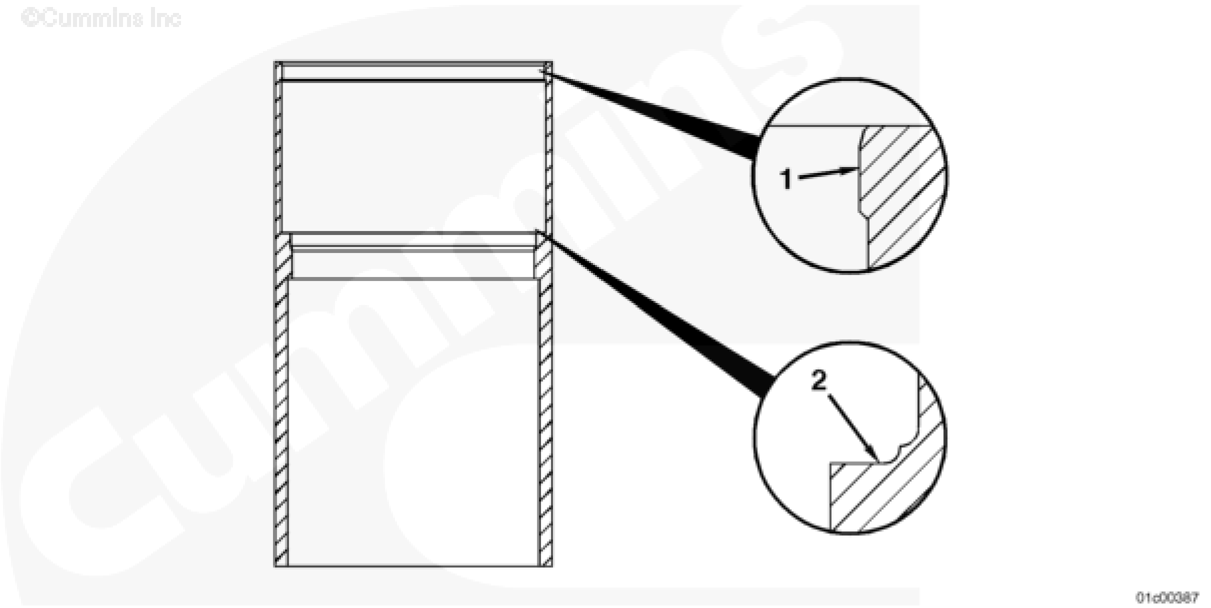
Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

Contents

This service/parts topic revises Procedure 001-026, Cylinder Block, for Signature, ISX, and QSX15 Series Engines.



Upper bore erosion (1) occurs where the cylinder liner upper o-ring comes in contact with the vertical surface of the block upper bore. It begins as corrosion; iron oxide embeds in the liner o-ring. The iron oxide is very abrasive and wears away the block. Corrosion is accelerated by high temperature and poorly maintained coolant chemistry. It should be pointed out that this erosion is **not** a failure and the block should be reused. Refer to Procedure 001-026 (/qs3/pubsys2/xml/en/procedures/10/10-001-026-tr.html), Cylinder Block.

The correct repair is to reinstall the liner or liners with new o-rings. The number and type of liner(s) used is dependent on the guidelines presently in place for the repair that is being conducted that resulted in the discovery of the upper bore erosion. Refer to Procedure 001-028

(/qs3/pubsys2/xml/en/procedures/10/10-001-028-tr.html), Cylinder Liner and Procedure 002-021 (/qs3/pubsys2/xml/en/procedures/10/10-002-021.html), Cylinder Head Gasket.

Liner ledge pitting (2) has an appearance very similar to liner pitting. Contributing factors to it's formation are: combustion gasses in the cooling system, poorly maintained coolant chemistry, and excessive load reversal at the liner to block contact area. Liner ledge pitting starts at the radius and progresses towards the center of the bore. In most circumstances this would **not** be a block failure.

To determine block reusability, several conditions **must** be met. First, you **must** be able to obtain correct liner protrusion. The number and type of liner(s) used is dependent on the guidelines presently in place for the repair that is being conducted that resulted in the discovery of the liner ledge pitting. Refer to Procedure 001-028 (/qs3/pubsys2/xml/en/procedures/10/10-001-028-tr.html), Cylinder Head Gasket. The second condition that **must** be met is that there can be no coolant leaks past the liner seat and the lower o-ring. Refer to Procedure 001-027 (/qs3/pubsys2/xml/en/procedures/10/10-001-027.html), Cylinder Block and Liner Seats.

Related Procedures

10-001-026 (/qs3/pubsys2/xml/en/procedures/10/10-001-026.html)	Cylinder Block (/qs3/pubsys2/xml/en/procedures/10/10-001-026.html)

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